

Case Study : Creating a DynamoDB Table with Car Details

Step 1: Sign in to the AWS Management Console

1. Open your web browser and go to the [AWS Management Console](#).
2. Sign in with your AWS credentials.

Step 2: Navigate to DynamoDB

1. In the AWS Management Console, type "DynamoDB" in the search bar and select DynamoDB from the drop-down list.

Step 3: Create a New Table

1. On the DynamoDB dashboard, click on the Create table button.
2. Enter the Table name. For this case study, let's name it CarDetails.
3. Define the Primary key:
 - Partition key: CarID (String)
 - Sort key: Model (String) (optional, if you want to include it)

Step 4: Configure Table Settings

1. Table settings:
 - Leave the default settings for Read/Write capacity mode as On-demand unless you have specific throughput requirements.
 - Optionally, you can enable Auto scaling if you choose Provisioned capacity mode.
2. Encryption:
 - Use the default AWS owned CMK (Customer Master Key) for encryption unless you have specific encryption requirements.

Step 5: Add Secondary Indexes (Optional)

1. If you need to query the table using attributes other than the primary key, you can add Global Secondary Indexes (GSI) or Local Secondary Indexes (LSI).
 - For example, you can add a GSI with Make as the partition key and Year as the sort key.

Step 6: Review and Create

1. Review all the settings you have configured.
2. Click on the Create table button.

Step 7: Add Items to the Table

1. Once the table is created, navigate to the Items tab.
2. Click on the Create item button.
3. Enter the details for a car. For example:
 - CarID: 1
 - Model: Model S
 - Make: Tesla
 - Year: 2022
 - Color: Red
 - Price: 79999
4. Click on the Save button to add the item.
5. Repeat the process to add more car details.

Step 8: Query and Scan the Table

1. To query the table, go to the Explore table items tab.
2. Use the Query or Scan options to retrieve data based on your requirements.

Step 9: Monitor and Manage the Table

1. Use the Metrics tab to monitor the table's performance.
2. Use the Alarms tab to set up alarms for monitoring read/write capacity and other metrics.
3. Use the Backups tab to create on-demand backups or enable point-in-time recovery.